# Mahmoud Alyosify

LLM & Generative AI Engineer | Fine-Tuning, RAG & Inference Optimisation
MSc Artificial Intelligence, Queen's University (Canada)
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## PROFESSIONAL SUMMARY

LLM & Generative AI Engineer specialising in **RAG, agentic LLM workflows, fine-tuning and inference-time optimisation** of open-weight language models. Currently an **AI Engineer Intern at Injaz Tech**, building an agentic RAG assistant with **LangChain, LangGraph, FAISS and FastAPI**. Completing an **MSc in Artificial Intelligence at Queen's University, Canada** (expected **Nov 2026**) on a **Digilians Presidential Scholarship**; master's research (**MinimalLM**) tests training-free decoding-time control of LLM verbosity on **Mistral, Llama-3.1 and Qwen2.5** against a strict truncation baseline. Hands-on with **LoRA/QLoRA fine-tuning at 159K-record scale**, self-hosted inference, LLM evaluation (BERTScore, ROUGE, LLM-as-Judge) and AWS GPU deployment. Focused on making language-model systems **cheaper, faster and measurably reliable** in production.

## EDUCATION

**MSc in Artificial Intelligence** — *Sep 2025 – Nov 2026*
*Queen's University, Kingston, Ontario, Canada* — Expected Nov 2026 · GPA: 4.30 / 4.30

- **Digilians Presidential Scholarship** — competitive national AI graduate fellowship.
- **Master's Research Project (CISC 898):** *MinimalLM: Verbosity-Aware Decoding for Efficient Language Model Generation* — Supervisor: Dr. Rahatara Ferdousi.
- Coursework: **Large Language Models, Generative AI, Deep Learning, Cloud Computing, End-to-End ML Pipelines**.

**B.Sc. Computer & Information Science (Bioinformatics)** — *Sep 2019 – Jul 2023*
*Assiut University, Egypt* — GPA: 3.53 / 4.00

- Graduation Project: **Vitalism Solution** — **Grade A+**; **Semi-Finalist, Microsoft Imagine Cup 2023**.

## TECHNICAL SKILLS

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|---|---|
| **LLM & Generative AI** | Large Language Models, Instruction Fine-Tuning, LoRA, QLoRA, PEFT, RAG, Agentic RAG (Routing, Relevance Grading, Query Rewriting, Self-Checks), Hybrid Search (BM25 + Dense, RRF), Cross-Encoder Re-Ranking, Custom Decoding Strategies, LogitsProcessors, Sampling & Entropy-Based Gating, Prompt Engineering, Structured Outputs, Tool / Function Calling, Multi-Agent Systems |
| **Frameworks & Serving** | PyTorch, Hugging Face Transformers, PEFT, Datasets, Accelerate, LangChain, LangGraph, Ollama, Pydantic, FAISS, ChromaDB, ONNX, FastAPI, Streamlit, Self-Hosted Inference, Docker, NVIDIA A40 GPU, AWS |
| **Models Worked With** | Mistral-7B-Instruct, Llama-3.1-8B-Instruct, Meta-Llama-3-8B, Qwen2.5-3B-Instruct |
| **LLM Evaluation** | MT-Bench, XSum, SQuAD-v2, BERTScore, ROUGE, Exact-Match / F1, Length-Controlled LLM-as-Judge, RAG Groundedness & Retrieval Metrics (Top-1, MRR), CI Evaluation Gates, Non-Inferiority Testing, Ablation Studies, Deterministic & Reproducible Benchmarking |
| **Efficiency & Systems** | Token-Cost Reduction, TTFT & Inter-Token Latency Profiling, GPU Energy Measurement (NVML), KV-Cache-Free Methods, Quantised Fine-Tuning (4-bit) |
| **Programming** | Python, SQL, C++, C#, JavaScript, Bash, Git/GitHub, Linux |

## PROFESSIONAL EXPERIENCE

**AI Engineer Intern** — *Oct 2026 – Present*
*Injaz Tech — Injaz Group (formerly Fingerprint Consultancy)* — Maadi, Cairo, Egypt

- Built an **agentic (corrective) RAG** assistant for internal-documentation Q&A with **LangChain** and **LangGraph**: question routing, **FAISS** retrieval with **cross-encoder re-ranking**, LLM relevance grading, query rewriting, hallucination self-checks, conversational memory and code-verified citations.
- Benchmarked chunking and retrieval strategies (dense, BM25, hybrid RRF, MMR, multi-query, re-ranking), raising top-1 retrieval accuracy from **75% to 94%**; redesigned the relevance grader as an evidence-quoting **Pydantic** structured output verified in code, lifting its recall from **17% to 88%** and cutting false refusals from **5/20 to 1/20**.
- Served it through **FastAPI** with SSE token streaming, a context-aware answer cache (repeat-question latency **34.7 s to 1.6 s**), retries and prompt-injection guardrails, backed by a **136-test pytest** suite and a CI evaluation gate over a 34-question LLM-as-judge benchmark.

**Part-Time Instructor — Machine Learning & AI for Business** — *Jul 2025 – Present*
*National Telecommunication Institute (NTI)* — Cairo, Arish & Assiut, Egypt

- Teach **Machine Learning** (90-hour bootcamps in Cairo, Arish and Assiut) and **AI for Business** modules in NTI's Summer Internships Program, on separate engagements.

**AI in Cybersecurity Intern** *Jun 2025 – Jul 2025*
*e& Egypt — via NTI (On-the-Job Training)* Cairo, Egypt

- Applied AI-driven tooling across threat intelligence, digital forensics and enterprise security analysis.

**Systems Engineer (Military Service)** *Feb 2024 – Mar 2025*
*National Company for Roads Building & Development* Cairo, Egypt

- Supported backend systems for national toll infrastructure serving **50%+ of Egyptian vehicles**; co-developed an internal data-management application in **C#, .NET, SQL Server**.

**Online Instructor — AI, Algorithms & Data Science** *2020 – Present*
*Udemy & YouTube* Remote

- Built Arabic-language technical curricula reaching **10,000+ learners** and **700K+ YouTube views**.

## SELECTED LLM RESEARCH & ENGINEERING PROJECTS

**MinimalLM: Verbosity-Aware Decoding for Efficient Language Model Generation** *2026 – Present*
*MSc Research Project (CISC 898), Queen's University · Supervisor: Dr. Rahatara Ferdousi*

- Research on **training-free decoding-time control** of LLM verbosity, tested against plain truncation on **Mistral-7B, Llama-3.1-8B and Qwen2.5-3B**. An entropy-gated **LogitsProcessor** cut **25.5–34.9%** of MT-Bench tokens (**19.9–34.2%** on a sealed test split) within pre-registered BERTScore margins, but an audit showed the saving was truncation.
- A survey of **23** training-free mechanism families found none that beat truncation without oracle labels. The proposed **Entailment-Licensed In-Decoding Elision** (sentence encoder + NLI model) beats truncation on every model, removing 0.9–2.2% of tokens; scoping it to discourse units cut reader-judged harmful elisions from 49–69% to **10.2%**.

**Horus-OSINT — Domain-Adapted LLM Assistant on AWS** *2026*
*Meta-Llama-3-8B · Instruction Fine-Tuning · AWS · Python*

- Fine-tuned **Meta-Llama-3-8B** on **159,826** curated GTD/GDELT records, building the full pipeline from dataset construction and instruction formatting through training to evaluation.
- Deployed the adapted model on **AWS** as an interactive assistant that answers natural-language questions over global event and threat-intelligence data.

**HORUS Sentinel — Multi-Agent LLM Platform over a Knowledge Graph** *2026*
*Multi-Agent Systems · Fine-Tuned LLM · Knowledge Graphs · Self-Hosted Inference*

- Architected a swarm of specialised agents that continuously collect open-source data and correlate every finding into a **living knowledge graph** used as the retrieval substrate for generation.
- A **self-hosted, fine-tuned language model** reasons over that graph to produce **prioritised, evidence-backed reports**, grounding every claim in retrieved source material.

**Automated Multimodal Agent — PDF to Narrated Presentation** *2026*
*Generative AI · Agentic Orchestration · Multimodal Generation*

- Built an autonomous think–plan–act workflow that converts PDFs into fully narrated PowerPoint decks by chaining document understanding, structured slide generation and audio synthesis.

**Additional Relevant Work**

- **Human vs. Agentic Pull Requests** — reproducible GitHub-mining study quantifying review effort and failure modes of LLM-generated versus human-authored code.
- **SimCLR-Vision-SSL** — contrastive representation learning with **ONNX/FAISS** embedding retrieval — transferable to RAG vector-search pipelines.

## CERTIFICATIONS & AWARDS

**Generative AI & Prompt Engineering** — IBM / Coursera

**Machine Learning Specialization** — DeepLearning.AI

**Digilians Presidential Scholarship** — Queen's University

**1st Place** — Smart Cities Hackathon 2022

**AWS Academy** — ML for NLP, ML Foundations, Data Engineering

**HCIP – AI** (Huawei)

**Semi-Finalist** — Microsoft Imagine Cup 2023

**Ideal Student Award** — Assiut University, 2022–2023

## LANGUAGES

- **Arabic:** Native · **English:** Professional Working Proficiency · **French:** Basic.